

How to search for data with BigQuery

As you have been learning, data analysis in BigQuery is the process of examining large datasets to uncover hidden patterns, correlations, and other insights. In this reading, you'll learn more about how BigQuery facilitates data analysis and how Dataplex can be used to find datasets and tables.

The power of BigQuery

Data discovery can be daunting with the sheer volume of information available. BigQuery transforms this challenge into an opportunity with extensive search capabilities. With the use of SQL commands, BigQuery allows for complex searches including pattern matching, exact phrases, and logical combinations to make the search for data as granular as you need.

Use Dataplex for enhanced data discovery

As datasets grow in size and complexity, it becomes increasingly important to organize and manage metadata. Proper metadata management ensures that data analysis is reliable and the insights derived are based on accurate and current data. Together with tools like Dataplex, you can manage large data catalogs to streamline the search process in BigQuery. Dataplex enhances BigQuery's capabilities by organizing, managing, and governing data across various environments. Dataplex automatically captures metadata, making it easy to locate specific datasets based on properties like content and usage. It simplifies the process of finding relevant data but also ensures the data is consistent, up-to-date, and governed according to security standards. You can leverage Dataplex within BigQuery to navigate through vast amounts of data with ease, making your data analysis tasks more effective and insightful.

Simplify data searches with BigQuery

Once you've identified datasets or tables you need, BigQuery's search function allows you to conduct text-based searches within a table, even if you're unfamiliar with its schema. This capability saves time and effort because you don't have to complete the search process manually to find unique data elements buried in unstructured text and semi-structured JSON.

BigQuery allows you to employ advanced search algorithms to query and search across entire tables or specific columns using keywords or phrases. BigQuery search functions reduce manual effort and optimize data retrieval, leading to faster and more relevant results, which is essential when working with large and complex datasets. You can search through unstructured or semistructured data with complex schemas like log entries or network activity.

Navigate BigQuery

To effectively use BigQuery's search capabilities, first navigate to the appropriate dataset and table within the BigQuery interface.

To access and verify a dataset, complete the following steps:

1. Navigate to public datasets
2. Select your dataset
3. Navigate to the desired table
4. Review the schema
5. Review the details section

Use a public dataset for practice

If you'd like, you can use BigQuery to follow along with this example. Practice with the public dataset `planet_features` to access geospatial data within the `geo_openstreetmap` dataset.

In your BigQuery sandbox, search "openstreetmap" to locate the `geo_openstreetmap` dataset. You'll see 16 different tables. Click on the `planet_features` table to review the schema and details. Notice there are over 900 million rows of data in this dataset.

The screenshot shows the Google Cloud BigQuery interface. In the Explorer on the left, a search for "openstreetmap" has been performed, showing results under the "bigquery-public-data" project, specifically within the "geo_openstreetmap" dataset. The "planet_features" table is selected. The main panel displays the "DETAILS" for this table.

Table Info

Table ID	bigquery-public-data.geo_openstreetmap.planet_features
Created	Mar 28, 2020, 9:22:05 AM UTC
Last modified	Nov 15, 2021, 10:13:21 AM UTC
Table expiration	NEVER
Data location	US
Default collation	
Description	The planet_features table was created by transforming the OSM planet.pbf file's contents of (way, node, relation) records into GDAL-type records (point, line, multilinestring, multipolygon, other_relation) using ogr2ogr. Software used to perform the ETL is located here: https://github.com/allenday/bigquery-openstreetmap/tree/master/osm2bigquery https://github.com/allenday/bigquery-openstreetmap/tree/master/bin

Storage info

Number of rows	993,320,998
Total logical bytes	327.75 GB
Active logical bytes	0 B
Long term logical bytes	327.75 GB
Total physical bytes	255.47 GB

At the bottom, there are tabs for "PERSONAL HISTORY" and "PROJECT HISTORY", and a "REFRESH" button.

Write queries with SEARCH()

Once you've navigated to your chosen dataset, here's how to construct a basic query using the **SEARCH()** function:

1. Specify what you want to see
2. Identify where to look for it
3. Define what you're looking for
4. Validate your search results

Here's a basic template for your search query:

```
Unset
SELECT *
FROM (
    SELECT *, SEARCH(column_to_search, 'search_term') AS
    search_result
FROM 'dataset.table'
)
WHERE search_result = TRUE;
```

The query template is useful for searching repeated **STRUCT** or **JSON** columns that are otherwise difficult to search and interpret.

Using the public dataset from earlier, search "all_tags" which is a repeated struct to locate all records with the tag **lake**.

```
Unset
SELECT *
FROM (
    SELECT *, SEARCH(all_tags, 'lake') AS search_result
FROM 'bigquery-public-data.geo_openstreetmap.plant_features'
)
WHERE search_result = TRUE;
```

Query results

SAVE RESULTS

EXPLORE DATA

JOB INFORMATION

RESULTS

JSON

EXECUTION DETAILS

osm_way_id	osm_version	osm_timestamp	all_tags	geometry
			2 highway trunk	
			3 lanes 4	
			4 maxspeed 50	
			5 name Airport Boulevard	

Key takeaways

BigQuery and Dataplex offer a powerful combination for data exploration, enabling users to uncover valuable insights efficiently. By understanding and using BigQuery's search functions and Dataplex's metadata organization, you can navigate the vast sea of data with ease, ensuring your data analysis is as effective as possible.